MODULE 3

MBRAMOS

Networking and Communications

IP Addressing & Binary Conversion

Decimal To Binary

To convert a decimal number to binary using the 128 64 32 16 8 4 2 1 table, you compare the number to these place values from left to right and decide whether each value "fits" into the number. If the number fits with your decimal, add one (1) beneath the table and if not, add zero (0). To build your table, start from the right and multiply by 2s in order, starting from number 1 until you reach 128.

128	64	32	16	8	4	2	1
X	X	X	X	X	X	X	X

10	1010
26	11010
38	100110

178	10110010
145	10010001
255	11111111

Binary To Decimal

To convert a binary number to decimal using the 128 64 32 16 8 4 2 1 table, start by writing the table. Add each binary digit by its corresponding place value (128, 64, 32, 16, 8, 4, 2, 1). If the digit is 1, take the full value; if it is 0, take 0. Finally, add up all the values obtained. The total sum is the decimal equivalent of the binary number. This time, from right to left.

128	64	32	16	8	4	2	1
x	x	x	x	x	x	x	x

101	5
1101	13
10010	18

111000	56
100111	39
1010101	85

Structure of an IP address

Now for the structure of an IP address, imagine each number in an IP address as an X. An IPv4 address contains four decimal numbers in total. Each X represents 8 bits, so an IP address consists of four Xs, equals to 32 bits in total. Each X, or decimal number, in an IP address is called an octet. As shown in binary/decimal conversion, the highest possible value in an octet is 255.

X	X	X	X	VALID?

8	34	92	17	YES
54	201	33	5	YES
151	101	1	69	YES
300	12	45	9	NO

Classes of IP Addresses

IP addresses in IPv4 were traditionally divided into classes based on how many bits are used for the network portion versus the host portion. To know which class an IP address is in, you compare it to the 1st Octet.

Class	1st Octet	Number of Hosts	Subnet Mask
Class A	1 - 126	16.7 Million	255.0.0.0
Class B	128 - 191	65 Thousand	255.255.0.0
Class C	192 - 223	254	255.255.255.0

IP ADDRESS	WHAT CLASS?
21.6.10.4	A
96.4.106.254	A
160.10.4.100	B
200.100.200.1	C

Loopback IP Address

If you can see the highlighted 1st Octet Column, there's no 127 Octet. It's because the number 127 is reserved for a loopback IP address. A loopback IP address is a special IP used by a device to test its own network interface without sending data over a physical network. In IPv4, the loopback range is 127.0.0.0 to 127.255.255.255, with 127.0.0.1 being the most commonly used. When you "ping" 127.0.0.1, the device sends the data to itself, which helps verify that the TCP/IP stack and network software are working correctly. Loopback addresses are internal and cannot be accessed by other devices on the network, making them useful for testing and troubleshooting.

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Class B	128 - 191	65 Thousand	255.255.0.0
Class C	192 - 223	254	255.255.255.0

What Are Private IP addresses

Private IP addresses are reserved for use within private networks, such as homes, offices, or enterprise LANs, and are not routable on the public internet. They allow devices within the same network to communicate without consuming public IP addresses.

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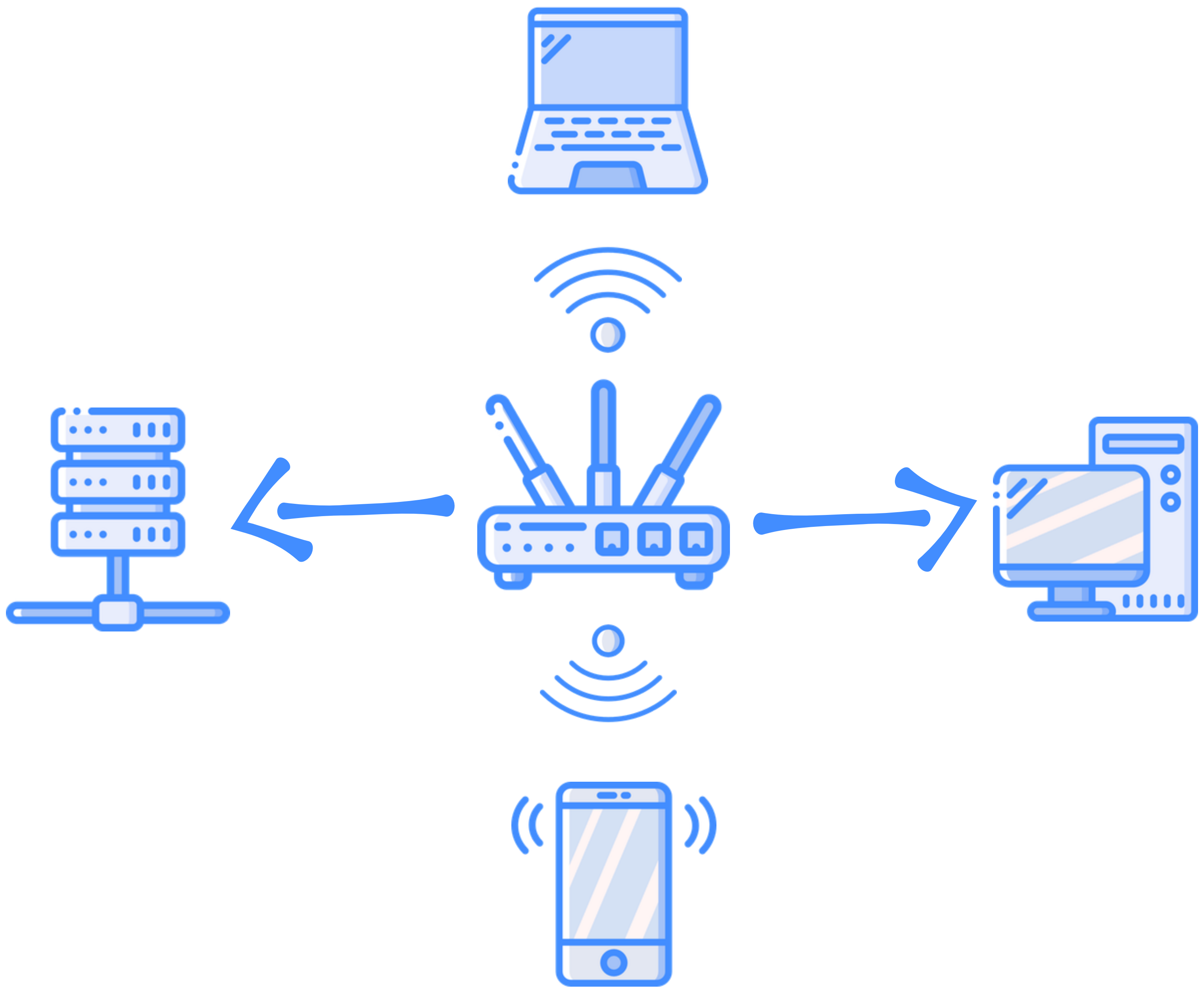
Class A	1 - 126	16.7 Million	255.0.0.0
Class B	128 - 191	65 Thousand	255.255.0.0
Class C	192 - 223	254	255.255.255.0

Non-routable IPv4 Addresses (Private/NAT Addresses)		
Class	First Address	Last Address
Class A	10.0.0.0	10.255.255.255
Class B	172.16.0.0	172.31.255.255
Class C	192.168.0.0	192.168.255.255

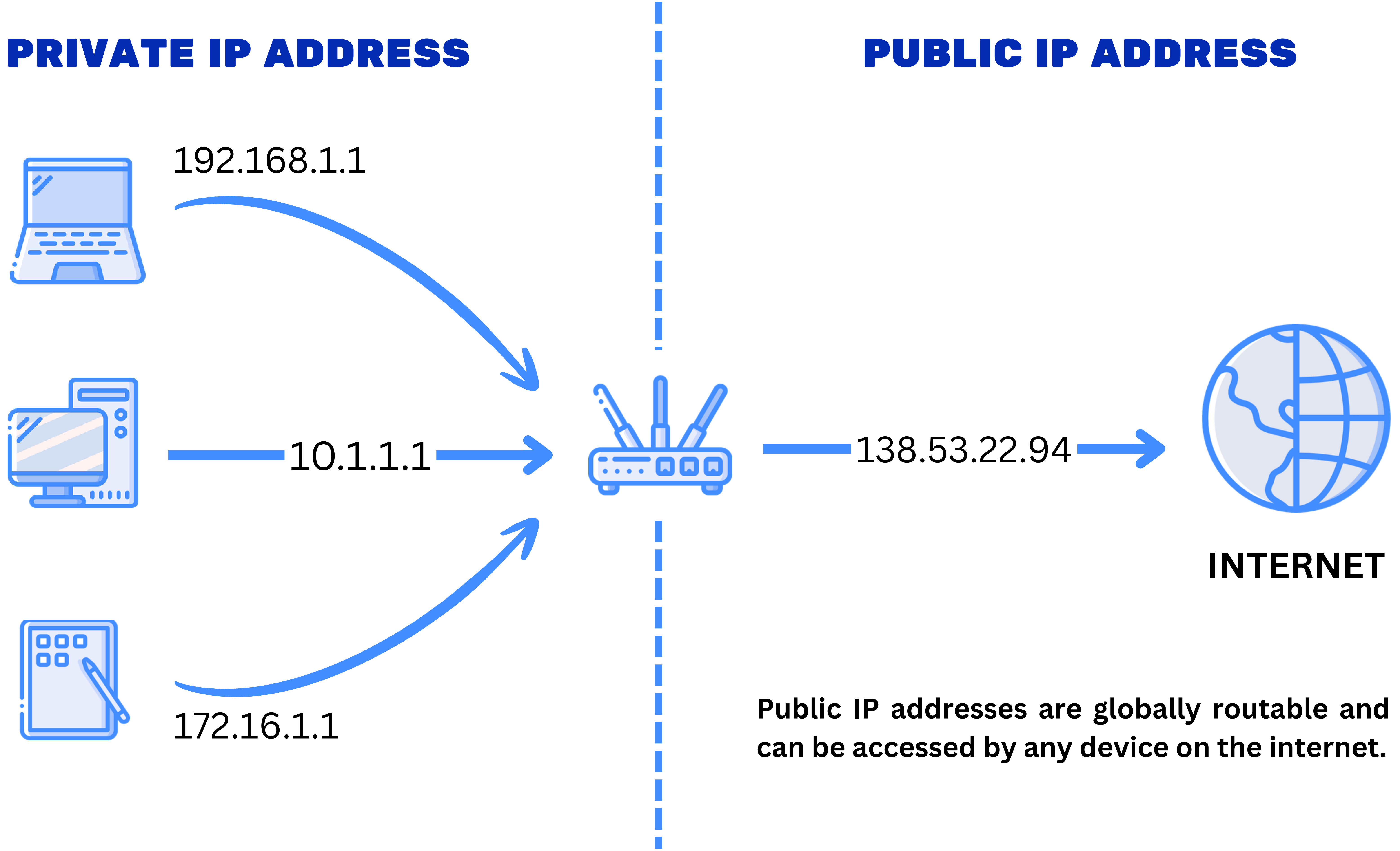
Where are Private IPs used?



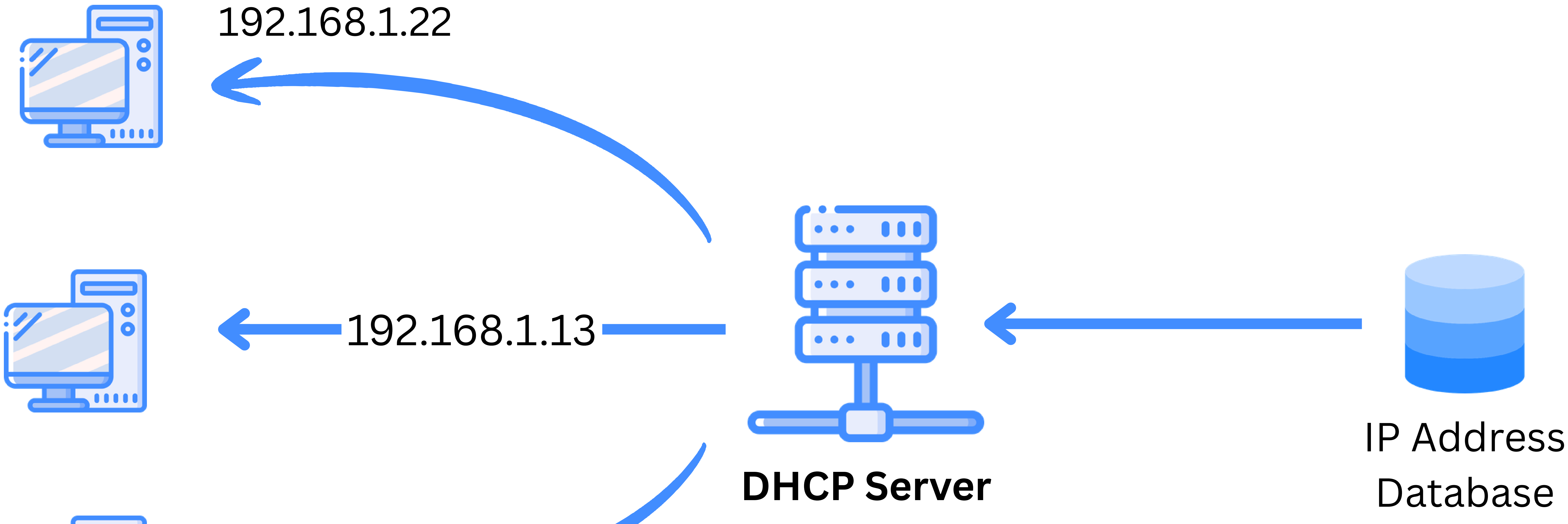
Private IPs in a LAN

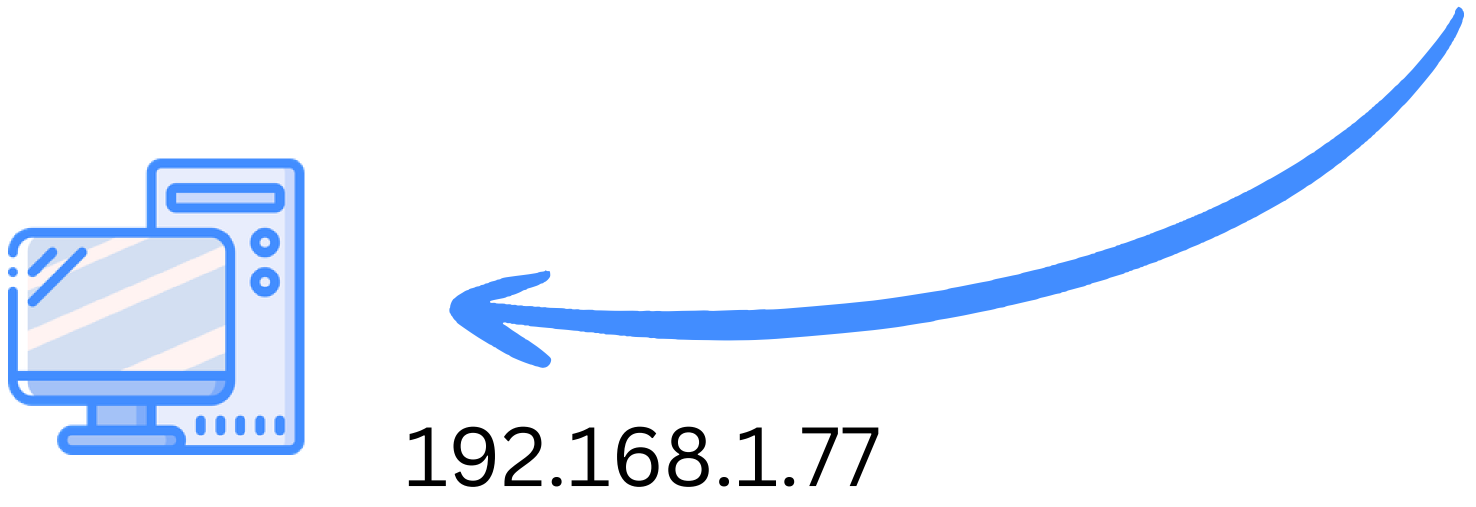
- Private IP addresses are specifically designed for use within local networks, such as a home, office, or enterprise LAN.
- <u>They cannot be accessed directly from the internet.</u>
- Examples of private IP ranges:
    - 10.0.0.0 - 10.255.255.255
    - 172.16.0.0 - 172.31.255.255
    - 192.168.0.0 - 192.168.255.255
- Devices in a LAN (computers, printers, phones) are almost always assigned private IPs, either manually or via DHCP.
- Since private IPs cannot directly access the Internet, the router uses NAT to map multiple private IPs in the LAN to its single public IP, allowing internal devices to communicate with the Internet.

NETWORK ADDRESS TRANSLATION (NAT)



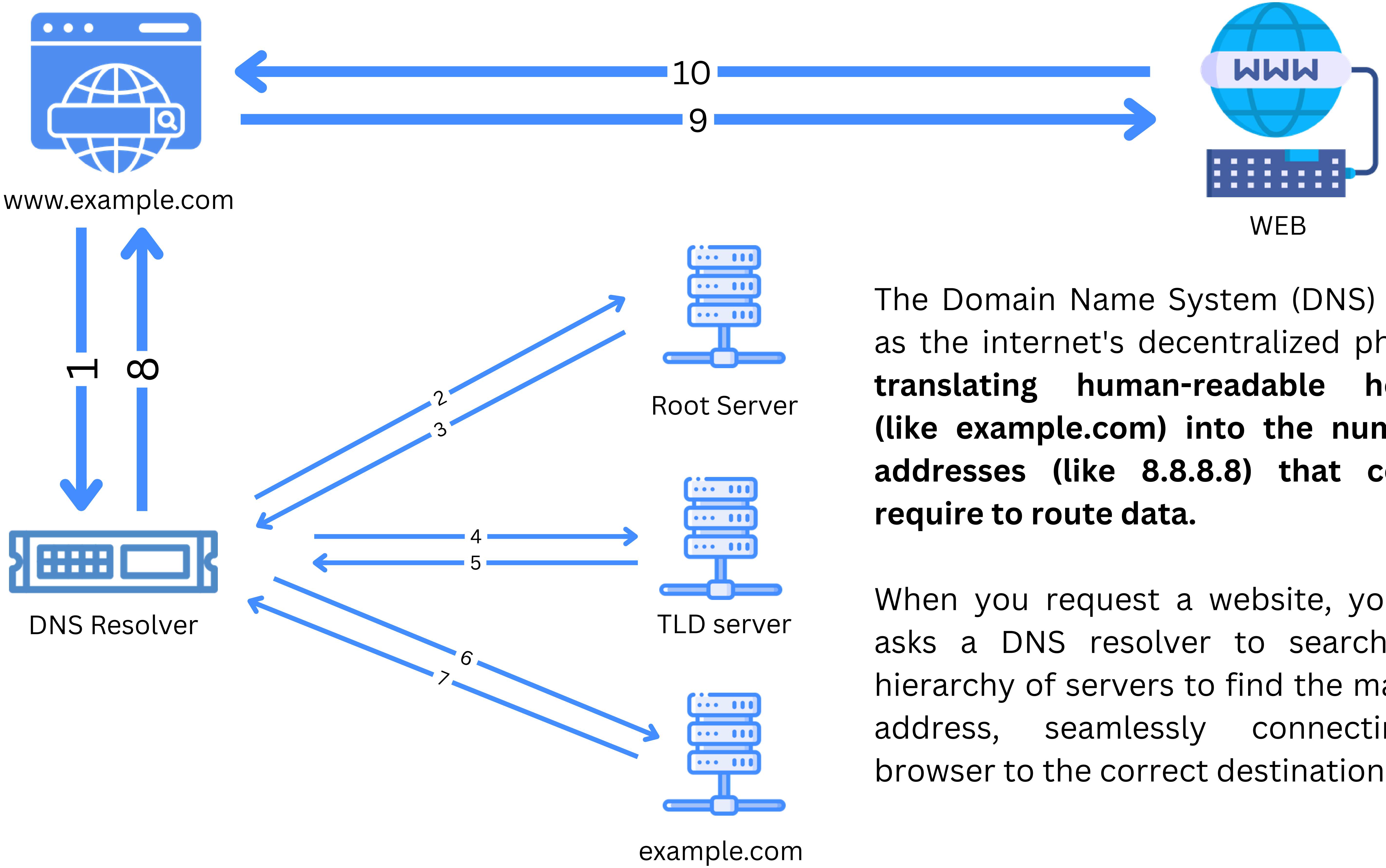
DYNAMIC HOST CONFIGURATION PROTOCOL (DHCP)





DHCP is an application-layer, client-server protocol that **automatically assigns IP addresses** and essential routing parameters—such as subnet masks, default gateways to network devices via UDP, enabling seamless IP communication without manual configuration.

DOMAIN NAME SYSTEM (DNS)



The Domain Name System (DNS) functions as the internet's decentralized phonebook, **translating human-readable hostnames (like example.com) into the numerical IP addresses (like 8.8.8.8) that computers require to route data.**

When you request a website, your device asks a DNS resolver to search a strict hierarchy of servers to find the matching IP address, seamlessly connecting your browser to the correct destination.